

IBM University Engagement Project Submission

“Exploring the Power of Agentic AI with IBM Granite and LangFlow”

Project Tile – Investment Analyst Agent

Domain of Project – Finance

Student Name	Recent Passport Photo	Email ID	Mobile number [WhatsApp]
Amey Patil		Ameywpatil@gmail.com	9403431809

Recommended font style and size:

Font Style: Arial

Font Size (Heading): 28

Font Size (Content): 20

Problem Statement -

Problem Statement : Investment Intelligence Agent

The Challenge

Good investment is the foundation of financial health, but individuals often struggle with **investment planning, portion control, and tracking investment intake**. Information about financial healthy investments, disease-specific investment, and market composition is scattered across unreliable sources. Without **personalized, real-time guidance**, many fail to maintain balanced investments or make informed market choices.

The Objective

Build an **AI-powered Investment Intelligence Agent** that delivers **personalized investmentary insights and guidance** for individuals, families, and financial health professionals. The solution should:

- **Personalized Investment Plans** – Recommend portfolios based on age, financial health conditions, allergies, cultural preferences, and fitness goals.
- **Market Data RAG Retrieval** – Fetch and summarize investmental values (prices, returns, volatility, and financial indicators) from reliable sources.
- **Investment Tracking & Feedback** – Allow users to log portfolios via text, images, or voice, and provide instant investmental analysis.
- **Preventive Financial Health Focus** – Suggest investments for risk management and portfolio diversification (e.g., diabetes-friendly or heart-financial healthy plans).
- **Multi-Agent System** –
 - **Investment Knowledge Agent** (fetches investmental data)
 - **Investment Recommendation Agent** (personalized portfolio plans)
 - **Financial Health Advisory Agent** (preventive financial health suggestions)
 - **Market Log & Feedback Agent** (portfolio tracking + analysis)
- **Visualization Dashboard** – Show daily/weekly nutrient intake, deficiencies, and progress toward financial health goals.

Proposed Solution -

Proposed Solution: AI-Powered Investment Intelligence Agent

The proposed solution is a **multi-agent AI-powered Investment Intelligence Agent** built using **LangFlow**, **IBM watsonx.ai**, and **Granite Models** to provide personalized, real-time investmentary guidance. The system addresses challenges in investment planning, investment tracking, and preventive financial healthcare through an intelligent, modular architecture.

At its core, the solution uses a **Retrieval-Augmented Generation (RAG)** pipeline to fetch accurate investmental data from trusted sources such as NASDAQ historical datasets and financial market datasets. This data is embedded and stored in a vector database, enabling the system to retrieve and summarize market investment details (prices, returns, volatility, and financial indicators) using Granite models.

The system consists of four key agents:

Investment Knowledge Agent retrieves and summarizes investmental information using RAG.

Investment Recommendation Agent generates personalized portfolio plans based on user inputs such as age, financial health conditions, investmentary preferences, and fitness goals.

Financial Health Advisory Agent provides preventive financial health suggestions, including disease-specific investments (e.g., diabetes-friendly or heart-financial healthy plans).

Market Log & Feedback Agent allows users to log portfolios via text, voice, or images and delivers instant investmental analysis with actionable feedback.

The solution includes a **user-friendly dashboard** (built with Streamlit or React) that visualizes portfolio performance and market exposure, highlights deficiencies, and tracks progress toward financial health goals.

LangFlow is used to orchestrate agent workflows, while watsonx.ai ensures scalable model deployment, embeddings, and AI governance. Granite models power reasoning, summarization, and personalization.

Overall, this solution delivers an **end-to-end intelligent investment assistant** that promotes financial healthier lifestyles through personalized insights, real-time feedback, and preventive financial healthcare recommendations.

Technology Used

- **IBM Granite model** - for eg – ibm-granite-3-2-8b – For natural language understanding, reasoning, and personalization
- **IBM Cloud** – It provides the infrastructure to build, deploy, and scale the Investment Intelligence Agent efficiently. IBM Cloud enables a **reliable, secure, and scalable environment** for deploying the Investment Intelligence Agent.
- **RAG**– RAG to **fetch real data first, then generate intelligent answers**, making the Investment Intelligence Agent more trustworthy and effective.
- **IBM Watsonx.ai** - For model deployment, embeddings, and AI governance

Langflow component Used -

- 1) **Chat Input** – Interface where users enter their queries, preferences, or portfolio details (text/voice/image).
- 2) **IBM watsonx AI Agent** - Processes user input using Granite models to generate personalized investment insights and recommendations.
- 3) **Name of IBM Granite model** - Granite-4.0-8B-Instruct → Fast, efficient for real-time investment recommendations
- 4) **Chat output** -Displays AI-generated responses such as investment plans, investmental analysis, and financial health suggestions.
- 5) **File component** - **NASDAQ CSV**.

Role of Agentic AI in the solution

Role of Agentic AI in Investment Intelligence Agent

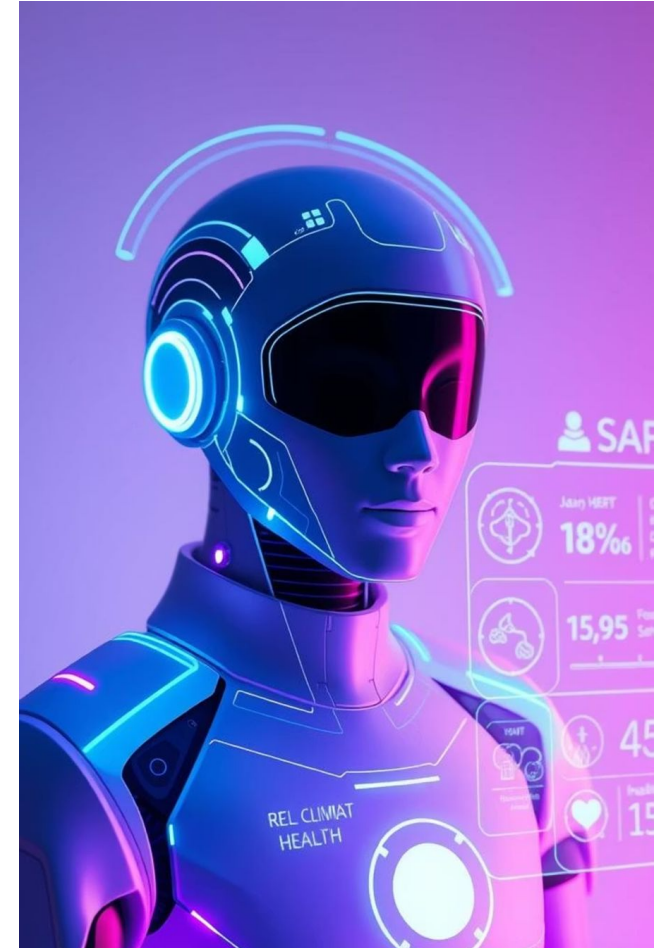
Agentic AI enables the Investment Intelligence Agent to function as an intelligent, autonomous system rather than a simple chatbot. Using **IBM watsonx.ai** and **IBM Granite models**, it coordinates multiple specialized agents to deliver personalized investment guidance.

Each agent performs a specific role—such as retrieving investmental data, generating investment plans, providing financial health advice, and analyzing market logs—while working collaboratively. This **multi-agent approach** ensures efficient task execution and accurate results.

Agentic AI also brings **context awareness**, understanding user details like age, financial health conditions, and fitness goals to generate tailored recommendations. It supports **real-time adaptation**, continuously updating suggestions based on user inputs like daily portfolios.

Additionally, agents communicate with each other, enabling better decision-making and improved outcomes.

Overall, Agentic AI makes the system proactive, personalized, and goal-driven, acting like a **smart digital investmentist** that helps users maintain a financial better investment decisions.



Novelty and Uniqueness

The proposed Investment Intelligence Agent stands out by combining **Agentic AI, RAG, and personalization** into a single intelligent system.

Multi-Agent Intelligence – Instead of a single chatbot, multiple specialized agents collaborate to deliver accurate and context-aware investment guidance.

RAG-Based Accuracy – Integrates real-time data retrieval with AI generation, reducing misinformation and ensuring trustworthy recommendations.

Hyper-Personalization – Tailors investment plans based on financial health conditions, cultural preferences, and fitness goals.

Real-Time Feedback Loop – Continuously adapts recommendations based on daily market logs and user behavior.

Multi-Modal Input – Supports text, voice, and image-based portfolio tracking for better usability.

Preventive Financial Healthcare Focus – Goes beyond investment planning to provide disease-specific investment advice.

Scalable & Enterprise-Ready – Built on **IBM watsonx.ai** with **IBM Granite models**, ensuring reliability and governance.

This makes the solution a **smart, adaptive, and proactive digital investmentist**, not just a static investment recommendation tool.

GitHub Link

Github Repo - <https://github.com/Tervicke/ibm-bootcamp>

Live Link - <https://tervicke.github.io/ibm-bootcamp/>

.

.

.

Future Scope

1.Integration with Wearables & Financial Health Devices –

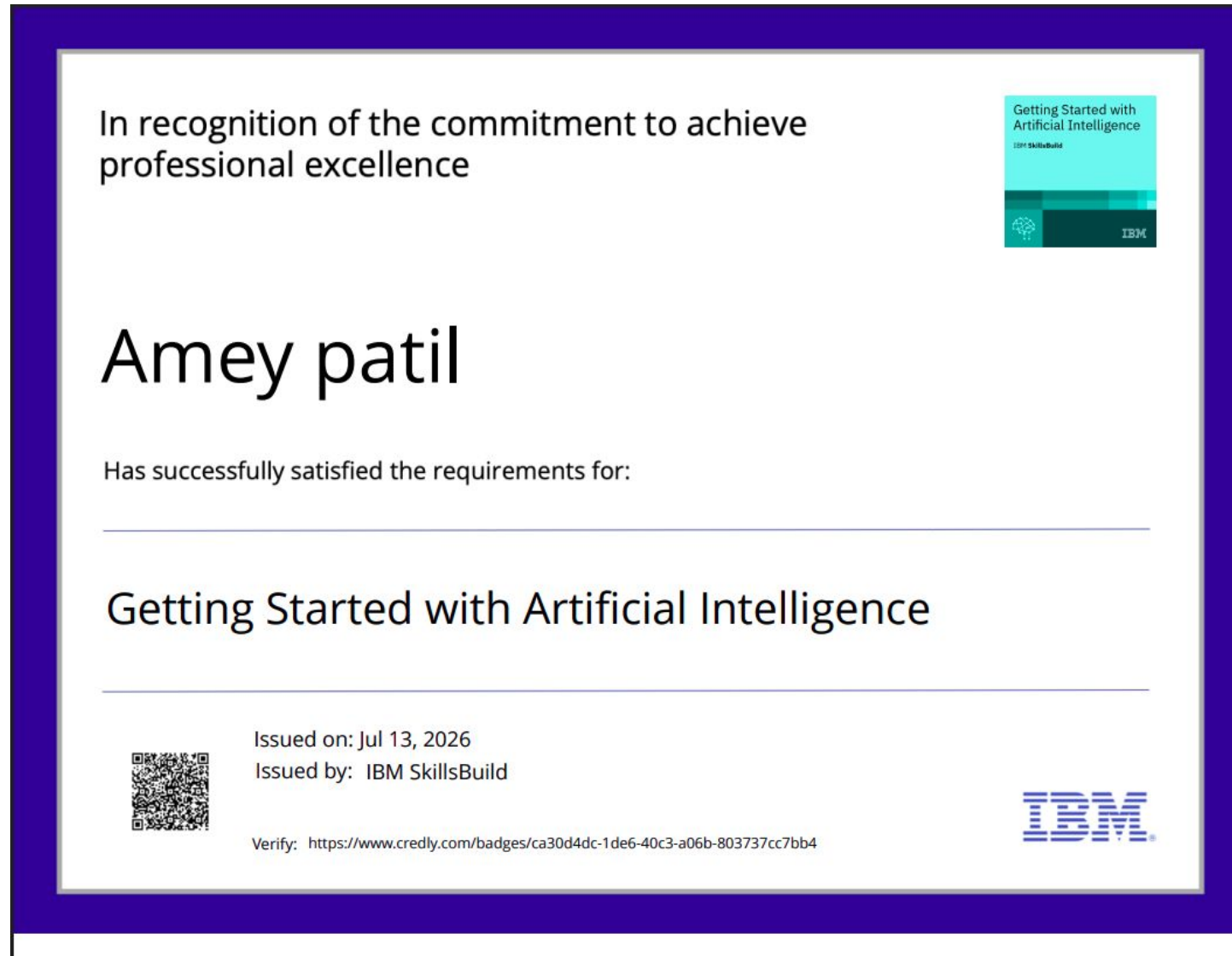
The system can be integrated with smartwatches and fitness trackers to collect real-time data such as steps, calories burned, heart rate, and sleep patterns. This will enable the Investment Intelligence Agent to provide more accurate and dynamic investment recommendations based on actual user activity and financial health metrics.

2.AI-Powered Voice Assistant & Multilingual Support –

Enhancing the system with a voice-enabled assistant and support for multiple languages will improve accessibility. Users can interact naturally in their preferred language, making the solution more inclusive and suitable for diverse populations, especially in regions with varied linguistic backgrounds.

Certificates Earned:

Please add your credly certificate



Certificates Earned:

Add your IBM BOB Certificate

IBM SkillsBuild	Completion Certificate
	
This certificate is presented to	
Amey patil	
for the completion of	
Lab: Troubleshoot Your Code Using IBM Bob	
(ALM-COURSE_4071307)	
According to the Adobe Learning Manager system of record	
Completion date: 14 Jul 2026 (GMT)	Learning hours: 30 mins

Thank You!

Thank you for your time and interest.